

COGNITIVE SCIENCE AND TECHNOLOGY: Building Futuristic Education Systems

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Internet boom and emergence of smart-phones as the largest-used devices to consume data have disrupted several industries like newspapers, book publishing, music and films, among others. Now, education joins that list, thanks to the emergence of massive open online courses, or MOOCs. The world of education is moving very fast from traditional classroom teaching towards MOOCs by using course portals like Coursera, FutureLearn, NPTEL, SWAYAM and so on. However, this shifting of the education paradigm is posing a number of challenges.

In traditional class-room teaching, a teacher can judge the learning level of students by their gestures and can adjust as per the learners' pace. But in online teaching, the instructor is unable to know on real-time basis whether the learners are actually understanding the contents. So, it has become necessary to introduce intelligent tutoring systems (ITSes), using which it can be assessed how deeply the students are thinking while learning.

This system should be able to anticipate whether the student is struggling while learning by providing timely alerts. It should be able to select an adequate pedagogy that can adapt to the learner's state and adjust its pace or mode of delivery by judging the learning curve beforehand instead of waiting for a test later.

To achieve this vision, knowledge of cognitive load of the learner can play a vital role. Cognitive load relates to the amount of information that the working memory can hold at any given time. Cognitive Load Theory (CLT) can be used to design learning materials, so that information can be presented at a pace and level of complexity that the learner can easily grasp and understand.

Researchers are working on this idea to build a system that can automatically measure cognitive load in real-time and optimise its instructional strategies to promote intrinsic

cognitive load while avoiding cognitive overload. To understand this deeply, let us understand the concept of CLT.

Cognitive Load Theory

CLT is built upon the widely-accepted model of human information processing. Three main parts to human information processing are sensory memory, working memory and long-term memory, as shown in Fig. 1.

Every day and everywhere we are bombarded with lots of sensory information. Although sensory memory filters out most of this information, it keeps an impression of the most important items that it later passes on to the working memory. Information from the sensory memory passes into the working memory, where it is either processed or discarded. Working memory can generally hold between five and nine items/chunks of information at any given time. This is central to CLT.

When the brain processes information, it categorises that information and moves it into long-term memory, where it is stored in knowledge structures called schemas. Concepts and habits are built on these schemas.

CLT was developed by John Sweller in the late 1980s. This theory suggests that learners can absorb and retain information effectively only if it is provided in such a way that it does not overload their mental capacity. The more information that is delivered at once, the more likely that the students will not actually learn what is being taught, nor will they be able to call upon that information for later use.

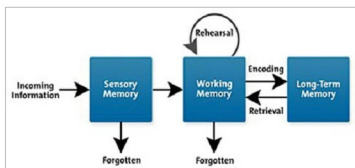


Fig. 1: Human information processing

Concerned with the scientific study of the human mind and its processes, cognitive science has evolved through the use of technologies that provide a window into the mind